



ADVANTAGES OF USING AI IN THE PUBLIC SECTOR



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Artificial Intelligence is becoming an important tool in government and public sector services because it can help agencies work faster, reduce errors, detect fraud, and improve services for citizens. Government agencies handle large amounts of data every day, including tax records, public benefits, emergency reports, healthcare information, and infrastructure data. AI can help analyze this information more quickly than humans alone. According to the U.S. Government Accountability Office, AI has the potential to support areas such as medical diagnosis, disaster forecasting, and national security. Still, it also creates concerns about bias, privacy, and oversight.

This report focuses on a real-world case study involving the U.S. Department of the Treasury's use of machine learning AI to detect fraud and improper payments. It also proposes an innovative AI application for local government services: an AI-powered citizen support and resource system.

Case Study Analysis: U.S. Treasury AI Fraud Detection

One strong example of AI in government is the U.S. Department of the Treasury's use of machine learning to detect fraud and improper payments. Fraud detection is a major public sector issue because government agencies distribute billions of dollars through programs such as tax refunds, benefits, grants, and emergency support. When fraudulent or improper payments occur, taxpayer money is wasted, and people who truly need assistance may experience delays.

In fiscal year 2024, the Treasury announced that its enhanced fraud detection processes, including machine learning AI, helped prevent and recover more than \$4 billion in fraud and improper payments. This was a major increase from \$652.7 million in fiscal year 2023. The problem being addressed was the need to detect suspicious payment activity more quickly and accurately, especially after fraud risks increased during and after the COVID-19 pandemic.

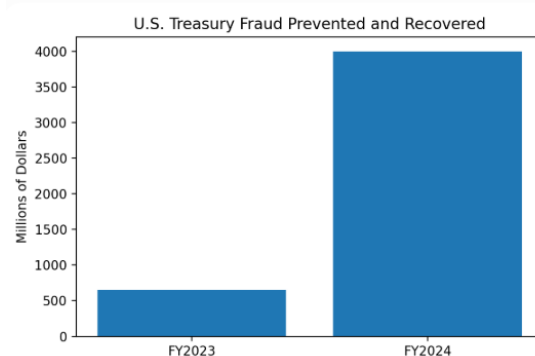


Figure 1. U.S. Treasury fraud was prevented and recovered through enhanced fraud detection and machine learning in FY2023 and FY2024.

The AI tools used in this case included machine learning systems that could review payment patterns, identify high-risk transactions, and prioritize suspicious cases for further review. Instead of relying only on manual review, AI helped scan large amounts of payment data and flag possible fraud. Treasury reported that risk-based screening helped prevent \$500 million, high-risk transaction identification helped prevent \$2.5 billion, and machine learning AI helped recover \$1 billion in Treasury check fraud.

The benefits of this case study are clear. First, AI improved efficiency by allowing the Treasury to review large volumes of payment data faster than traditional methods. Second, AI helped protect taxpayer money by identifying suspicious activity before or after payments were issued. Third, it allowed government workers to focus more attention on the highest-risk cases instead of spending time reviewing every transaction manually. This shows how AI can support human decision-making rather than completely replace human workers.

Beyond improving efficiency, AI also allows government agencies to make more informed decisions by identifying patterns that may not be obvious through manual review alone. Machine learning systems continuously improve as they analyze additional data, allowing them to recognize new fraud schemes and adapt to changing tactics used by criminals. This capability is especially valuable because fraud methods continue to evolve. AI can also help agencies prioritize investigations by assigning risk scores to suspicious transactions, ensuring that the most serious cases receive immediate attention. Rather than replacing government employees, AI serves as a decision-support tool that provides analysts with valuable insights while leaving the final decision to trained professionals. This partnership between technology and human expertise improves accuracy, reduces workloads, and enables agencies to respond more quickly to emerging threats while maintaining accountability and public trust.

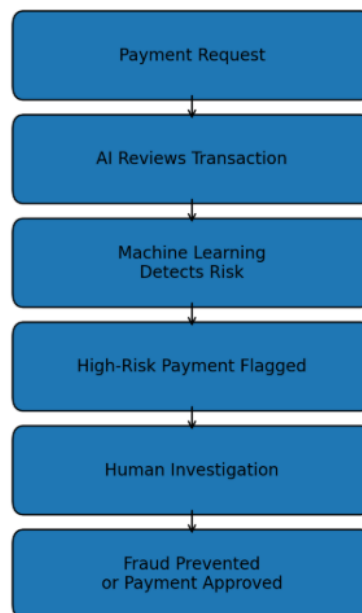


Figure 2. Simplified workflow of AI-assisted fraud detection used to identify high-risk government payments.

Another benefit is improved accountability. Government agencies are responsible for making sure public money is used correctly. By using AI to detect improper payments, the Treasury strengthened payment integrity and improved public trust. This is especially important because citizens expect government agencies to be careful with taxpayer funds.

However, there are also challenges and limitations. AI systems depend on data quality. If the data is incomplete, outdated, or biased, the system may make inaccurate predictions. There is also a risk of false positives, where legitimate payments are flagged as suspicious. This could delay payments for people who need them. Privacy is another concern because fraud detection systems may involve sensitive financial and personal information. The government must make sure AI systems are secure, transparent, and properly monitored.

The GAO has also noted that federal agencies face challenges when using AI, including policy compliance, technical resources, budget limitations, and keeping AI policies updated as the technology changes. These challenges show that AI should not be used without rules, oversight, and human review.

Table 1. Advantages and challenges of using AI in government and public sector services.

Advantages	Challenges
Faster processing	Data privacy concerns
Fraud detection	Algorithm bias
Better decision support	False positives
Saves taxpayer money	Cybersecurity risks
Improves citizen services	Human oversight required

Innovative Proposal: AI-Powered Citizen Support and Resource System

An innovative AI application for the public sector would be an AI-powered citizen support and resource system for local governments. Many citizens struggle to find the right public services because government websites can be confusing. People may need help with housing assistance, food support, unemployment services, public transportation, permits, emergency alerts, or healthcare resources. An AI system could act as a virtual public service assistant that helps residents find information and connect with the right department.



Figure 3. Proposed AI-powered citizen support system for improving access to government services.

This system could use natural language processing, enabling AI to understand questions in everyday language. For example, a citizen could type, “I need help paying rent,” and the AI could guide them to local housing assistance programs, eligibility requirements, deadlines, and contact information. It could also be available in multiple languages, making public services more accessible to people who do not speak English as their first language.

The expected outcomes would include faster service, reduced call center wait times, better access to public programs, and improved citizen satisfaction. Government employees would also benefit because the AI could answer common questions, allowing staff to focus on more complex cases. This would be especially useful during emergencies, when many people need quick information at the same time.

The system could also help local governments identify service gaps. For example, if many residents ask about food assistance in one neighborhood, the city could use that information to improve outreach or add resources in that area. This would help government leaders make better decisions based on real community needs.

An additional advantage of this proposed system is its ability to improve communication between citizens and local governments. Many residents are unaware of the programs and services available to them simply because the information is spread across multiple websites or is difficult to locate. An AI-powered assistant could provide personalized guidance based on a person's questions while remaining available 24 hours a day, seven days a week. During emergencies such as hurricanes, floods, or public health crises, the system could deliver real-time updates, evacuation information, shelter locations, and emergency assistance resources to

affected communities. It could also collect anonymous data about the most common requests, helping government officials identify trends and allocate resources where they are needed most. By combining AI with human oversight, local governments could provide faster, more accessible, and more responsive services while improving the overall experience for citizens seeking assistance.

There are challenges to this proposal. The AI must protect personal information and avoid giving incorrect advice. It should clearly explain that it is an assistant, not a final decision-maker. Human workers should still review sensitive cases, such as benefit eligibility or emergency assistance. The system should also be tested for bias to make sure it serves all communities fairly. The GAO's AI accountability framework emphasizes governance, data quality, performance, and monitoring, which would be important for this type of system.

Conclusion

AI offers many advantages for government and public sector services. The U.S. Treasury case study shows that machine learning can help detect fraud, recover public funds, and improve payment integrity. AI can process large amounts of data quickly and help government employees focus on the most important cases. At the same time, AI must be used carefully because of risks related to privacy, bias, data quality, and public trust.

The proposed AI-powered citizen support system would help local governments improve access to public services by giving residents faster and clearer guidance. If designed responsibly, AI can make government more efficient, responsive, and citizen-centered. Overall, AI should be viewed as a support tool that strengthens human decision-making, not a replacement for public servants.

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